

# MELDA KAYATEKIN

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## EDUCATION

### Stony Brook University — B.E. Biomedical Engineering

Expected May 2028

Presidential Scholarship | Dean's List

### The Bronx High School of Science — High School Diploma

June 2025

GPA: 4.0 / 94.77% | Terra Competition Semifinalist, Manne Research Institute (2023–2025) | Model UN Co-Chair, led 100+ delegates across weekly committee sessions

## RESEARCH & ENGINEERING EXPERIENCE

### Undergraduate Researcher — Eric Josephs Lab, Stony Brook University

Fall 2026 – Present

- Design and execute molecular bioengineering experiments supporting research on CRISPR based gene editing and gene therapy engineering, contributing to work focused on improving the safety, efficacy, and accessibility of gene therapies
- Maintain standardized protocols and technical documentation for DNA repair and gene editing assays to ensure reproducibility, building fluency in the rigorous documentation practices used throughout regulated medtech and biotech R&D

### Prototyping Fellow — Tikkun Olam Makers (TOM), New York

May 2026 – Present

- Design and prototype assistive medical devices that target hyperspecific needs of individuals with disabilities, working directly with end users to translate their unique physical and functional challenges into custom, purpose built solutions
- Apply iterative engineering design principles from initial concept through functional build, refining prototypes based on direct user feedback and real world testing
- Author technical documentation and replication manuals that enable open source distribution of adapted devices, expanding patient access to assistive technology

### Team Lead, VIP BEAR — Stony Brook University

Dec 2025 – Present

#### Bioengineering Education, Application and Research

- Oversee all team projects, working directly with United Cerebral Palsy to identify unmet mobility needs and guiding the team in developing specific engineering solutions for individuals with disabilities
- Translate clinical mobility needs identified by physical and occupational therapists into engineering design requirements for motorization, steering, and safety controls
- Develop standardized device manuals and collaborate directly with clinicians to validate that design adaptations meet real patient needs

### Engineering Workshop Assistant — CEAS, Stony Brook University

Aug 2026 – Present

- Assist students with hands on engineering projects during workshops, providing guidance on CAD modeling, 3D printing, and prototyping workflows
- Troubleshoot design and fabrication issues in real time, helping students move from digital models to functional printed parts

### Research Intern — Manne Research Institute

Summer 2023 – Fall 2024

- Studied tardigrade survival adaptations, investigating how sustained long term freeze stress affects physiological resilience and cellular survival mechanisms
- Conducted literature reviews and developed optimized lab protocols for biological sample processing
- Analyzed quantitative and qualitative data in Excel to identify trends and produce written technical reports

## PROJECTS

### Implantable Microfluidic Biosensor for Early Alzheimer's Detection — BME 100, Stony Brook

Fall 2025

#### University

- Designed a conceptual microfluidic diagnostic device using enzyme coated electrochemical sensors to detect Alzheimer's biomarkers (A $\beta$ -40, A $\beta$ -42, p-tau) in cerebrospinal fluid, with wireless transmission to a smartphone or smartwatch
- Earned faculty honorable mention for design innovation

### Interactive Image Editor, MATLAB App Designer

Fall 2026

- Built a GUI based image editing tool enabling real time RGB channel, brightness, and rotation adjustments, demonstrating digital image processing fundamentals and pixel level data manipulation

### 3D Printed Braille Keyboard with Audio Feedback (in progress)

2026

- Designing a Braille embossed keyboard from scratch in AutoCAD, currently integrating a voice sensor channel to audibly announce each keystroke for users who are blind or visually impaired

### Motorized Rotating Countertop System (in progress)

2026

- Designing a button activated, motorized rotating tray that circulates items from the back of a countertop to the front, improving kitchen accessibility for wheelchair users with limited reach

### Remote Physical Board Game Interface (in progress)

2026

- Developing a magnet and gantry system beneath a physical board game, controlled via Arduino and connected to an online game overlay integrated with Overjoyed, an accessible gaming software, enabling users with dexterity impairments to play physical board games remotely through a digital interface

## LEADERSHIP

- Engineering Fellow, Generation Teach** (Summer 2026) — designed and delivered a project based STEM curriculum introducing the engineering design process to young adults
- Engineering Peer Mentor, CEAS** (2026–Present) — mentors incoming CEAS students on course selection and academic transition
- Center Assistant, Astoria Kumon** (Feb 2025–Present) — provides 1-on-1 academic support in math and reading

## TECHNICAL SKILLS

**Programming:** Python (Beginner), Java, C++, MATLAB, GML

**Design & Prototyping:** SolidWorks, AutoCAD, 3D printing, iterative prototyping, technical documentation, assistive device design

**Engineering & Fabrication:** Arduino/embedded systems, circuit design, motor control, microfluidic device design, digital image processing

**Lab Skills:** Literature review, protocol development, biological sample processing, data analysis

**Tools:** Excel, Microsoft Office Suite, Google Workspace, Canva